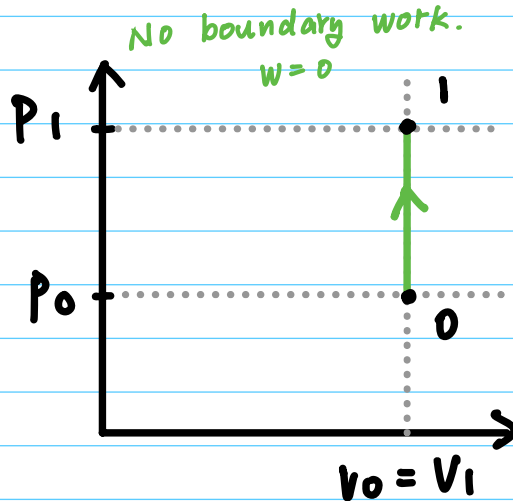


- Q1. A 0.8-m<sup>3</sup> rigid tank contains refrigerant-134a initially at 180 kPa and 45 percent quality. Heat is now transferred to the refrigerant until the pressure reaches 600 kPa. Determine (a) the mass of the refrigerant in the tank and (b) the amount of heat transferred. Also, show the process on a P-v diagram with respect to saturation lines.

Rigid tank.  $V = 0.8 \text{ m}^3 \Rightarrow$  constant volume  $R-134a \Rightarrow$  Table A-12



Energy balance:  $\Delta E_{\text{system}} = Q - W = \Delta U$   
&  $w = 0 \Rightarrow Q = \Delta U = m \cdot \Delta u$

Initial state:  $P_0 = 180 \text{ kPa}$ ,  $x = 0.45$

Final state:  $P_1 = 600 \text{ kPa}$ .

TABLE A-12

Saturated refrigerant-134a—Pressure table

|                            |                                              | Specific volume,<br>m <sup>3</sup> /kg   |                                         | Internal energy,<br>kJ/kg                |                                  | Enthalpy,<br>kJ/kg                      |                                          |                                  | Entropy,<br>kJ/kg·K                     |                                          |                                  |                                         |
|----------------------------|----------------------------------------------|------------------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|
| Press.,<br><i>P</i><br>kPa | Sat.<br>temp.,<br><i>T</i> <sub>sat</sub> °C | Sat.<br>liquid,<br><i>v</i> <sub>f</sub> | Sat.<br>vapor,<br><i>v</i> <sub>g</sub> | Sat.<br>liquid,<br><i>u</i> <sub>f</sub> | Evap.,<br><i>u</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>u</i> <sub>g</sub> | Sat.<br>liquid,<br><i>h</i> <sub>f</sub> | Evap.,<br><i>h</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>h</i> <sub>g</sub> | Sat.<br>liquid,<br><i>s</i> <sub>f</sub> | Evap.,<br><i>s</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>s</i> <sub>g</sub> |
| 60                         | −36.95                                       | 0.0007097                                | 0.31108                                 | 3.795                                    | 205.34                           | 209.13                                  | 3.837                                    | 223.96                           | 227.80                                  | 0.01633                                  | 0.94812                          | 0.96445                                 |
| 70                         | −33.87                                       | 0.0007143                                | 0.26921                                 | 7.672                                    | 203.23                           | 210.90                                  | 7.722                                    | 222.02                           | 229.74                                  | 0.03264                                  | 0.92783                          | 0.96047                                 |
| 80                         | −31.13                                       | 0.0007184                                | 0.23749                                 | 11.14                                    | 201.33                           | 212.48                                  | 11.20                                    | 220.27                           | 231.47                                  | 0.04707                                  | 0.91009                          | 0.95716                                 |
| 90                         | −28.65                                       | 0.0007222                                | 0.21261                                 | 14.30                                    | 199.60                           | 213.90                                  | 14.36                                    | 218.67                           | 233.04                                  | 0.06003                                  | 0.89431                          | 0.95434                                 |
| 100                        | −26.37                                       | 0.0007258                                | 0.19255                                 | 17.19                                    | 198.01                           | 215.21                                  | 17.27                                    | 217.19                           | 234.46                                  | 0.07182                                  | 0.88008                          | 0.95191                                 |
| 120                        | −22.32                                       | 0.0007323                                | 0.16216                                 | 22.38                                    | 195.15                           | 217.53                                  | 22.47                                    | 214.52                           | 236.99                                  | 0.09269                                  | 0.85520                          | 0.94789                                 |
| 140                        | −18.77                                       | 0.0007381                                | 0.14020                                 | 26.96                                    | 192.60                           | 219.56                                  | 27.06                                    | 212.13                           | 239.19                                  | 0.11080                                  | 0.83387                          | 0.94467                                 |
| 160                        | −15.60                                       | 0.0007435                                | 0.12355                                 | 31.06                                    | 190.31                           | 221.37                                  | 31.18                                    | 209.96                           | 241.14                                  | 0.12686                                  | 0.81517                          | 0.94202                                 |
| 180                        | −12.73                                       | 0.0007485                                | 0.11049                                 | 34.81                                    | 188.20                           | 223.01                                  | 34.94                                    | 207.95                           | 242.90                                  | 0.14131                                  | 0.79848                          | 0.93979                                 |
| 200                        | −10.09                                       | 0.0007532                                | 0.099951                                | 38.26                                    | 186.25                           | 224.51                                  | 38.41                                    | 206.09                           | 244.50                                  | 0.15449                                  | 0.78339                          | 0.93788                                 |
| 240                        | −5.38                                        | 0.0007618                                | 0.083983                                | 44.46                                    | 182.71                           | 227.17                                  | 44.64                                    | 202.68                           | 247.32                                  | 0.17786                                  | 0.75689                          | 0.93475                                 |
| 280                        | −1.25                                        | 0.0007697                                | 0.072434                                | 49.95                                    | 179.54                           | 229.49                                  | 50.16                                    | 199.61                           | 249.77                                  | 0.19822                                  | 0.73406                          | 0.93228                                 |
| 320                        | 2.46                                         | 0.0007771                                | 0.063681                                | 54.90                                    | 176.65                           | 231.55                                  | 55.14                                    | 196.78                           | 251.93                                  | 0.21631                                  | 0.71395                          | 0.93026                                 |
| 360                        | 5.82                                         | 0.0007840                                | 0.056809                                | 59.42                                    | 173.99                           | 233.41                                  | 59.70                                    | 194.15                           | 253.86                                  | 0.23265                                  | 0.69591                          | 0.92856                                 |
| 400                        | 8.91                                         | 0.0007905                                | 0.051266                                | 63.61                                    | 171.49                           | 235.10                                  | 63.92                                    | 191.68                           | 255.61                                  | 0.24757                                  | 0.67954                          | 0.92711                                 |
| 450                        | 12.46                                        | 0.0007983                                | 0.045677                                | 68.44                                    | 168.58                           | 237.03                                  | 68.80                                    | 188.78                           | 257.58                                  | 0.26462                                  | 0.66093                          | 0.92555                                 |
| 500                        | 15.71                                        | 0.0008058                                | 0.041168                                | 72.92                                    | 165.86                           | 238.77                                  | 73.32                                    | 186.04                           | 259.36                                  | 0.28021                                  | 0.64399                          | 0.92420                                 |
| 550                        | 18.73                                        | 0.0008129                                | 0.037452                                | 77.09                                    | 163.29                           | 240.38                                  | 77.54                                    | 183.44                           | 260.98                                  | 0.29460                                  | 0.62842                          | 0.92302                                 |
| 600                        | 21.55                                        | 0.0008198                                | 0.034335                                | 81.01                                    | 160.84                           | 241.86                                  | 81.50                                    | 180.95                           | 262.46                                  | 0.30799                                  | 0.61398                          | 0.92196                                 |
| 650                        | 24.20                                        | 0.0008265                                | 0.031680                                | 84.72                                    | 158.51                           | 243.23                                  | 85.26                                    | 178.56                           | 263.82                                  | 0.32052                                  | 0.60048                          | 0.92100                                 |
| 700                        | 26.69                                        | 0.0008331                                | 0.029392                                | 88.24                                    | 156.27                           | 244.51                                  | 88.82                                    | 176.26                           | 265.08                                  | 0.33232                                  | 0.58780                          | 0.92012                                 |
| 750                        | 29.06                                        | 0.0008395                                | 0.027398                                | 91.59                                    | 154.11                           | 245.70                                  | 92.22                                    | 174.03                           | 266.25                                  | 0.34348                                  | 0.57582                          | 0.91930                                 |
| 800                        | 31.31                                        | 0.0008457                                | 0.025645                                | 94.80                                    | 152.02                           | 246.82                                  | 95.48                                    | 171.86                           | 267.34                                  | 0.35408                                  | 0.56445                          | 0.91853                                 |
| 850                        | 33.45                                        | 0.0008519                                | 0.024091                                | 97.88                                    | 150.00                           | 247.88                                  | 98.61                                    | 169.75                           | 268.36                                  | 0.36417                                  | 0.55362                          | 0.91779                                 |
| 900                        | 35.51                                        | 0.0008580                                | 0.022703                                | 100.84                                   | 148.03                           | 248.88                                  | 101.62                                   | 167.69                           | 269.31                                  | 0.37383                                  | 0.54326                          | 0.91709                                 |
| 950                        | 37.48                                        | 0.0008640                                | 0.021456                                | 103.70                                   | 146.11                           | 249.82                                  | 104.52                                   | 165.68                           | 270.20                                  | 0.38307                                  | 0.53333                          | 0.91641                                 |
| 1000                       | 39.37                                        | 0.0008700                                | 0.020329                                | 106.47                                   | 144.24                           | 250.71                                  | 107.34                                   | 163.70                           | 271.04                                  | 0.39196                                  | 0.52378                          | 0.91574                                 |
| 1200                       | 46.29                                        | 0.0008935                                | 0.016728                                | 116.72                                   | 137.12                           | 253.84                                  | 117.79                                   | 156.12                           | 273.92                                  | 0.42449                                  | 0.48870                          | 0.91320                                 |
| 1400                       | 52.40                                        | 0.0009167                                | 0.014119                                | 125.96                                   | 130.44                           | 256.40                                  | 127.25                                   | 148.92                           | 276.17                                  | 0.45325                                  | 0.45742                          | 0.91067                                 |
| 1600                       | 57.88                                        | 0.0009400                                | 0.012134                                | 134.45                                   | 124.05                           | 258.50                                  | 135.96                                   | 141.96                           | 277.92                                  | 0.47921                                  | 0.42881                          | 0.90802                                 |
| 1800                       | 62.87                                        | 0.0009639                                | 0.010568                                | 142.36                                   | 117.85                           | 260.21                                  | 144.09                                   | 135.14                           | 279.23                                  | 0.50304                                  | 0.40213                          | 0.90517                                 |
| 2000                       | 67.45                                        | 0.0009887                                | 0.009297                                | 149.81                                   | 111.75                           | 261.56                                  | 151.78                                   | 128.36                           | 280.15                                  | 0.52519                                  | 0.37684                          | 0.90204                                 |
| 2500                       | 77.54                                        | 0.0010567                                | 0.006941                                | 167.02                                   | 96.47                            | 263.49                                  | 169.66                                   | 111.18                           | 280.84                                  | 0.57542                                  | 0.31701                          | 0.89243                                 |
| 3000                       | 86.16                                        | 0.0011410                                | 0.005272                                | 183.09                                   | 80.17                            | 263.26                                  | 186.51                                   | 92.57                            | 279.08                                  | 0.62133                                  | 0.25759                          | 0.87893                                 |

Initial state:  $\left\{ \begin{array}{l} v_f = 0.0007485 \text{ m}^3/\text{kg} \\ v_g = 0.11049 \text{ m}^3/\text{kg} \\ x = 0.45 \end{array} \right. \quad \& \quad \left\{ \begin{array}{l} u_f = 34.81 \text{ kJ/kg} \\ u_{fg} = 188.2 \text{ kJ/kg} \\ x = 0.45 \end{array} \right.$

$$\Rightarrow v_0 = x v_g + (1-x) v_f$$

$$= 0.45 \times 0.11049 + 0.55 \times 0.0007485 = 0.050132175 \text{ m}^3/\text{kg}.$$

$$m = \frac{V}{v_0} = \frac{0.8}{0.050132175} = 15.96 \text{ kg}.$$

$$u_0 = u_f + x \cdot u_{fg} = 34.81 + 0.45 \times 188.2 = 119.5 \text{ kJ/kg}$$

Final state } constant  $V. \Rightarrow v_{\text{final}} = v_0 = 0.05132175 \text{ m}^3/\text{kg}.$   
 $p_{\text{final}} = 600 \text{ kPa}.$   $> v_g (p = 600 \text{ kPa})$

superheated.  $\Rightarrow$  Table A-13.

| Superheated refrigerant-134a (Continued) |                                                         |            |            |              |                                                         |            |            |              |                                                         |            |            |              |  |
|------------------------------------------|---------------------------------------------------------|------------|------------|--------------|---------------------------------------------------------|------------|------------|--------------|---------------------------------------------------------|------------|------------|--------------|--|
| T<br>°C                                  | P = 0.50 MPa ( $T_{\text{sat}} = 15.71^\circ\text{C}$ ) |            |            |              | P = 0.60 MPa ( $T_{\text{sat}} = 21.55^\circ\text{C}$ ) |            |            |              | P = 0.70 MPa ( $T_{\text{sat}} = 26.69^\circ\text{C}$ ) |            |            |              |  |
|                                          | v<br>m <sup>3</sup> /kg                                 | u<br>kJ/kg | h<br>kJ/kg | s<br>kJ/kg·K | v<br>m <sup>3</sup> /kg                                 | u<br>kJ/kg | h<br>kJ/kg | s<br>kJ/kg·K | v<br>m <sup>3</sup> /kg                                 | u<br>kJ/kg | h<br>kJ/kg | s<br>kJ/kg·K |  |
| Sat.                                     | 0.041168                                                | 238.77     | 259.36     | 0.9242       | 0.034335                                                | 241.86     | 262.46     | 0.9220       | 0.029392                                                | 244.51     | 265.08     | 0.9201       |  |
| 20                                       | 0.042115                                                | 242.42     | 263.48     | 0.9384       |                                                         |            |            |              |                                                         |            |            |              |  |
| 30                                       | 0.044338                                                | 250.86     | 273.03     | 0.9704       | 0.035984                                                | 249.24     | 270.83     | 0.9500       | 0.029966                                                | 247.49     | 268.47     | 0.9314       |  |
| 40                                       | 0.046456                                                | 259.27     | 282.50     | 1.0011       | 0.037865                                                | 257.88     | 280.60     | 0.9817       | 0.031696                                                | 256.41     | 278.59     | 0.9642       |  |
| 50                                       | 0.048499                                                | 267.73     | 291.98     | 1.0309       | 0.039659                                                | 266.50     | 290.30     | 1.0122       | 0.033322                                                | 265.22     | 288.54     | 0.9955       |  |
| 60                                       | 0.050485                                                | 276.27     | 301.51     | 1.0600       | 0.041389                                                | 275.17     | 300.00     | 1.0417       | 0.034875                                                | 274.03     | 298.44     | 1.0257       |  |
| 70                                       | 0.052427                                                | 284.91     | 311.12     | 1.0884       | 0.043069                                                | 283.91     | 309.75     | 1.0706       | 0.036373                                                | 282.88     | 308.34     | 1.0550       |  |
| 80                                       | 0.054331                                                | 293.65     | 320.82     | 1.1163       | 0.044710                                                | 292.74     | 319.57     | 1.0988       | 0.037829                                                | 291.81     | 318.29     | 1.0835       |  |
| 90                                       | 0.056205                                                | 302.52     | 330.63     | 1.1436       | 0.046318                                                | 301.69     | 329.48     | 1.1265       | 0.039250                                                | 300.84     | 328.31     | 1.1115       |  |
| 100                                      | 0.058053                                                | 311.52     | 340.55     | 1.1706       | 0.047900                                                | 310.75     | 339.49     | 1.1536       | 0.040642                                                | 309.96     | 338.41     | 1.1389       |  |
| 110                                      | 0.059880                                                | 320.65     | 350.59     | 1.1971       | 0.049458                                                | 319.93     | 349.61     | 1.1804       | 0.042010                                                | 319.21     | 348.61     | 1.1659       |  |
| 120                                      | 0.061687                                                | 329.91     | 360.75     | 1.2233       | 0.050997                                                | 329.24     | 359.84     | 1.2068       | 0.043358                                                | 328.57     | 358.92     | 1.1925       |  |
| 130                                      | 0.063479                                                | 339.31     | 371.05     | 1.2492       | 0.052519                                                | 338.69     | 370.20     | 1.2328       | 0.044688                                                | 338.06     | 369.34     | 1.2186       |  |
| 140                                      | 0.065256                                                | 348.85     | 381.47     | 1.2747       | 0.054027                                                | 348.26     | 380.68     | 1.2585       | 0.046004                                                | 347.67     | 379.88     | 1.2445       |  |
| 150                                      | 0.067021                                                | 358.52     | 392.04     | 1.3000       | 0.055522                                                | 357.98     | 391.29     | 1.2838       | 0.047306                                                | 357.42     | 390.54     | 1.2700       |  |
| 160                                      | 0.068775                                                | 368.34     | 402.73     | 1.3250       | 0.057006                                                | 367.83     | 402.03     | 1.3089       | 0.048597                                                | 367.31     | 401.32     | 1.2952       |  |

$\Rightarrow$  At 0.6 MPa.

$$v_1 = 0.050997 \text{ m}^3/\text{kg}$$

$$v_2 = 0.052519 \text{ m}^3/\text{kg}$$

$$v_{\text{final}} = 0.05132175 \text{ m}^3/\text{kg}$$

$$u_1 = 329.24 \text{ kJ/kg}$$

$$u_2 = 338.69 \text{ kJ/kg}$$

$$\Rightarrow \text{Interpolation: } \frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} \Rightarrow \frac{v_{\text{final}} - v_1}{v_2 - v_1} = \frac{u_{\text{final}} - u_1}{u_2 - u_1}$$

$$\Rightarrow u_{\text{final}} = \frac{(v_{\text{final}} - v_1)(u_2 - u_1)}{v_2 - v_1} + u_1$$

$$= \frac{(0.05132175 - 0.050997)(338.69 - 329.24)}{0.052519 - 0.050997} + 329.24$$

$$= 327.2236482 \text{ kJ/kg} \approx 327.2236 \text{ kJ/kg}.$$

Heat transferred.  $Q = \Delta U = m \cdot \Delta u$

$$= m (u_{\text{final}} - u_0)$$

$$= 15.96 (327.2236 - 119.5)$$

$$\approx 3315 \text{ kJ}$$

Q2.

Steam at 95 kPa and 5 percent quality is contained in a spring-loaded piston-cylinder device, with an initial volume of 1.2 m<sup>3</sup>. Steam is now heated until its volume is 5.5 m<sup>3</sup> and its pressure is 230 kPa. Determine the heat transferred to and the work produced by the steam during this process.

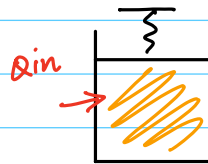
spring loaded.  $x = 0.05$ .

Initial:



$$P_0 = 95 \text{ kPa. } V_0 = 1.2 \text{ m}^3.$$

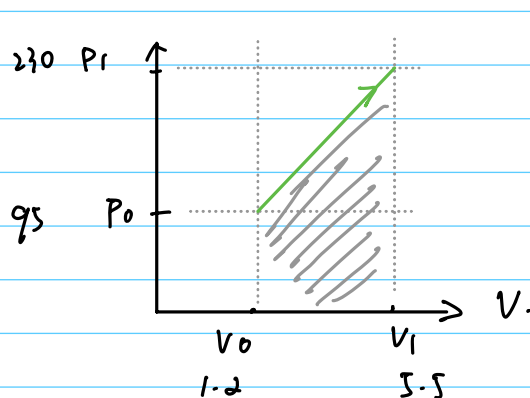
Final:



$$P_1 = 230 \text{ kPa. } V_1 = 5.5 \text{ m}^3.$$

⇒ P-V diagram.

spring load ⇒ Linear change.



$$\begin{aligned} W &= \frac{1}{2} (P_0 + P_1) (V_1 - V_0) \\ &= \frac{1}{2} (230 + 95) (5.5 - 1.2) \\ &= 698.75 \text{ kJ.} \end{aligned}$$

$$Q - W = \Delta U$$

V.

$$\Delta U = m \cdot \Delta u.$$

⇒ Find m.

⇒ Find  $\Delta U$ . Pressure table A-5.

| Press.,<br>P, kPa | Sat. temp.,<br>T <sub>sat</sub> , °C | Specific volume,<br>m <sup>3</sup> /kg |                               | Internal energy,<br>kJ/kg      |                           |                               | Enthalpy,<br>kJ/kg             |                           |                               | Entropy,<br>kJ/kg·K            |                           |                               |
|-------------------|--------------------------------------|----------------------------------------|-------------------------------|--------------------------------|---------------------------|-------------------------------|--------------------------------|---------------------------|-------------------------------|--------------------------------|---------------------------|-------------------------------|
|                   |                                      | Sat. liquid,<br>u <sub>f</sub>         | Sat. vapor,<br>u <sub>g</sub> | Sat. liquid,<br>u <sub>f</sub> | Evap.,<br>u <sub>fg</sub> | Sat. vapor,<br>u <sub>g</sub> | Sat. liquid,<br>h <sub>f</sub> | Evap.,<br>h <sub>fg</sub> | Sat. vapor,<br>h <sub>g</sub> | Sat. liquid,<br>s <sub>f</sub> | Evap.,<br>s <sub>fg</sub> | Sat. vapor,<br>s <sub>g</sub> |
| 1.0               | 6.97                                 | 0.001000                               | 129.19                        | 29.302                         | 2355.2                    | 2384.5                        | 29.303                         | 2484.4                    | 2513.7                        | 0.1059                         | 8.8690                    | 8.9749                        |
| 1.5               | 13.02                                | 0.001001                               | 87.964                        | 54.686                         | 2338.1                    | 2392.8                        | 54.688                         | 2470.1                    | 2524.7                        | 0.1956                         | 8.6314                    | 8.8270                        |
| 2.0               | 17.50                                | 0.001001                               | 66.990                        | 73.431                         | 2325.5                    | 2398.9                        | 73.433                         | 2459.5                    | 2532.9                        | 0.2606                         | 8.4621                    | 8.7227                        |
| 2.5               | 21.08                                | 0.001002                               | 54.242                        | 88.422                         | 2315.4                    | 2403.8                        | 88.424                         | 2451.0                    | 2539.4                        | 0.3118                         | 8.3302                    | 8.6421                        |
| 3.0               | 24.08                                | 0.001003                               | 45.654                        | 100.98                         | 2306.9                    | 2407.9                        | 100.98                         | 2443.9                    | 2544.8                        | 0.3543                         | 8.2222                    | 8.5765                        |
| 4.0               | 28.96                                | 0.001004                               | 34.791                        | 121.39                         | 2293.1                    | 2414.5                        | 121.39                         | 2432.3                    | 2553.7                        | 0.4224                         | 8.0510                    | 8.4734                        |
| 5.0               | 32.87                                | 0.001005                               | 28.185                        | 137.75                         | 2282.1                    | 2419.8                        | 137.75                         | 2423.0                    | 2560.7                        | 0.4762                         | 7.9176                    | 8.3938                        |
| 7.5               | 40.29                                | 0.001008                               | 19.233                        | 168.74                         | 2261.1                    | 2429.8                        | 168.75                         | 2405.3                    | 2574.0                        | 0.5763                         | 7.6738                    | 8.2501                        |
| 10                | 45.81                                | 0.001010                               | 14.670                        | 191.79                         | 2245.4                    | 2437.2                        | 191.81                         | 2392.1                    | 2583.9                        | 0.6492                         | 7.4996                    | 8.1488                        |
| 15                | 53.97                                | 0.001014                               | 10.020                        | 225.93                         | 2222.1                    | 2448.0                        | 225.94                         | 2372.3                    | 2598.3                        | 0.7549                         | 7.2522                    | 8.0071                        |
| 20                | 60.06                                | 0.001017                               | 7.6481                        | 251.40                         | 2204.6                    | 2456.0                        | 251.42                         | 2357.5                    | 2608.9                        | 0.8320                         | 7.0752                    | 7.9073                        |
| 25                | 64.96                                | 0.001020                               | 6.2034                        | 271.93                         | 2190.4                    | 2462.4                        | 271.96                         | 2345.5                    | 2617.5                        | 0.8932                         | 6.9370                    | 7.8302                        |
| 30                | 69.09                                | 0.001022                               | 5.2287                        | 289.24                         | 2178.5                    | 2467.7                        | 289.27                         | 2335.3                    | 2624.6                        | 0.9441                         | 6.8234                    | 7.7675                        |
| 40                | 75.86                                | 0.001026                               | 3.9933                        | 317.58                         | 2158.8                    | 2476.3                        | 317.62                         | 2318.4                    | 2636.1                        | 1.0261                         | 6.6430                    | 7.6691                        |
| 50                | 81.32                                | 0.001030                               | 3.2403                        | 340.49                         | 2142.7                    | 2483.2                        | 340.54                         | 2304.7                    | 2645.2                        | 1.0912                         | 6.5019                    | 7.5931                        |
| 75                | 91.76                                | 0.001037                               | 2.2172                        | 384.36                         | 2111.8                    | 2496.1                        | 384.44                         | 2278.0                    | 2662.4                        | 1.2132                         | 6.2426                    | 7.4558                        |
| 100               | 99.61                                | 0.001043                               | 1.6941                        | 417.40                         | 2088.2                    | 2505.6                        | 417.51                         | 2257.5                    | 2675.0                        | 1.3028                         | 6.0562                    | 7.3589                        |
| 101.325           | 99.97                                | 0.001043                               | 1.6734                        | 418.95                         | 2087.0                    | 2506.0                        | 419.06                         | 2256.5                    | 2675.6                        | 1.3069                         | 6.0476                    | 7.3545                        |
| 125               | 105.97                               | 0.001048                               | 1.3750                        | 444.23                         | 2068.8                    | 2513.0                        | 444.36                         | 2240.6                    | 2684.9                        | 1.3741                         | 5.9100                    | 7.2841                        |
| 150               | 111.35                               | 0.001053                               | 1.1594                        | 466.97                         | 2052.3                    | 2519.2                        | 467.13                         | 2226.0                    | 2693.1                        | 1.4337                         | 5.7894                    | 7.2231                        |
| 175               | 116.04                               | 0.001057                               | 1.0037                        | 486.82                         | 2037.7                    | 2524.5                        | 487.01                         | 2213.1                    | 2700.2                        | 1.4850                         | 5.6865                    | 7.1716                        |
| 200               | 120.21                               | 0.001061                               | 0.88578                       | 504.50                         | 2024.6                    | 2529.1                        | 504.71                         | 2201.6                    | 2706.3                        | 1.5302                         | 5.5968                    | 7.1270                        |
| 225               | 123.97                               | 0.001064                               | 0.79329                       | 520.47                         | 2012.7                    | 2533.2                        | 520.71                         | 2191.0                    | 2711.7                        | 1.5706                         | 5.5171                    | 7.0877                        |
| 250               | 127.41                               | 0.001067                               | 0.71873                       | 535.08                         | 2001.8                    | 2536.8                        | 535.35                         | 2181.2                    | 2716.5                        | 1.6072                         | 5.4453                    | 7.0525                        |
| 275               | 130.58                               | 0.001070                               | 0.65732                       | 548.57                         | 1991.6                    | 2540.1                        | 548.86                         | 2172.0                    | 2720.9                        | 1.6408                         | 5.3800                    | 7.0207                        |
| 300               | 133.52                               | 0.001073                               | 0.60582                       | 561.11                         | 1982.1                    | 2543.0                        | 561.43                         | 2163.5                    | 2724.9                        | 1.6717                         | 5.3200                    | 6.9917                        |
| 325               | 136.27                               | 0.001076                               | 0.56199                       | 572.84                         | 1973.1                    | 2545.5                        | 573.19                         | 2155.4                    | 2728.6                        | 1.7005                         | 5.2645                    | 6.9650                        |
| 350               | 138.86                               | 0.001079                               | 0.52422                       | 583.89                         | 1964.6                    | 2548.5                        | 584.26                         | 2147.7                    | 2732.0                        | 1.7274                         | 5.2128                    | 6.9402                        |
| 375               | 141.30                               | 0.001082                               | 0.49132                       | 594.73                         | 1956.1                    | 2551.0                        | 595.11                         | 2139.9                    | 2735.0                        | 1.7526                         | 5.1656                    | 6.9161                        |

$$P_0 = 95 \text{ kPa.}$$

$$P_1 = 230 \text{ kPa.}$$

State I:

$$\begin{aligned} y &= \frac{(P - P_1) \cdot (v_2 - v_1)}{(P_2 - P_1)} + v_1 \\ &= \frac{(195 - 75) \cdot (0.001043 - 0.001037)}{100 - 75} + 0.001037 \\ &= 0.0010418 \text{ m}^3/\text{kg.} \end{aligned}$$

$$v_g = \frac{(195 - 75) \cdot (1.6941 - 2.2172)}{100 - 75} + 2.2172 = 1.78696 \text{ m}^3/\text{kg}$$

$$\text{Given } x = 0.05. \Rightarrow v = x \cdot v_g + (1 - x) \cdot v_f.$$

$$= 0.05 \times 1.78696 + (1 - 0.05) \times 0.0010418 = 0.09033771 \text{ m}^3/\text{kg.}$$

$$\Rightarrow m = \frac{V}{v} = \frac{1.2}{0.0903771} = 13.2835 \text{ kg} \approx 13.28 \text{ kg.}$$

Internal energy.  $\left\{ \begin{array}{l} u_f = \frac{(195-75)(417.4-384.36)}{100-75} + 384.36 = 410.792 \text{ kJ/kg.} \\ u_g = \frac{(195-75)(2505.6-2496.1)}{100-75} + 2496.1 = 2503.7 \text{ kJ/kg.} \end{array} \right.$

$$u = x \cdot u_g + (1-x) \cdot u_f = 0.05 \times 2503.7 + (1-0.05) \times 410.792 = 513.4374 \text{ kJ/kg.} \approx 513.44 \text{ kJ/kg}$$

Find state.  $v_1 = \frac{V_1}{m} = \frac{5.5}{13.2835} \approx 0.4140 \text{ m}^3/\text{kg}$

$$P = 230 \text{ kPa.}$$

**TABLE A-5**  
Saturated water—Pressure table

| Press.,<br>P kPa | Sat. temp.,<br>T <sub>sat</sub> °C | Specific volume,<br>m <sup>3</sup> /kg |                               |                               | Internal energy,<br>kJ/kg      |                           |                               | Enthalpy,<br>kJ/kg             |                           |                               | Entropy,<br>kJ/kg·K            |                           |                               |
|------------------|------------------------------------|----------------------------------------|-------------------------------|-------------------------------|--------------------------------|---------------------------|-------------------------------|--------------------------------|---------------------------|-------------------------------|--------------------------------|---------------------------|-------------------------------|
|                  |                                    | Sat. liquid,<br>v <sub>f</sub>         | Sat. vapor,<br>v <sub>g</sub> | Sat. vapor,<br>v <sub>g</sub> | Sat. liquid,<br>u <sub>f</sub> | Evap.,<br>u <sub>fg</sub> | Sat. vapor,<br>u <sub>g</sub> | Sat. liquid,<br>h <sub>f</sub> | Evap.,<br>h <sub>fg</sub> | Sat. vapor,<br>h <sub>g</sub> | Sat. liquid,<br>s <sub>f</sub> | Evap.,<br>s <sub>fg</sub> | Sat. vapor,<br>s <sub>g</sub> |
| 1.0              | 6.97                               | 0.001000                               | 129.19                        |                               | 29.302                         | 2355.2                    | 2384.5                        | 29.303                         | 2484.4                    | 2513.7                        | 0.1059                         | 8.8690                    | 8.9749                        |
| 1.5              | 13.02                              | 0.001001                               | 87.964                        |                               | 54.686                         | 2338.1                    | 2392.8                        | 54.688                         | 2470.1                    | 2524.7                        | 0.1956                         | 8.6314                    | 8.8270                        |
| 2.0              | 17.50                              | 0.001001                               | 66.990                        |                               | 73.431                         | 2325.5                    | 2398.9                        | 73.433                         | 2459.5                    | 2532.9                        | 0.2606                         | 8.4621                    | 8.7227                        |
| 2.5              | 21.08                              | 0.001002                               | 54.242                        |                               | 88.422                         | 2315.4                    | 2403.8                        | 88.424                         | 2451.0                    | 2539.4                        | 0.3118                         | 8.3302                    | 8.6421                        |
| 3.0              | 24.08                              | 0.001003                               | 45.654                        |                               | 100.98                         | 2306.9                    | 2407.9                        | 100.98                         | 2443.9                    | 2544.8                        | 0.3543                         | 8.2222                    | 8.5765                        |
| 4.0              | 28.96                              | 0.001004                               | 34.701                        |                               | 121.39                         | 2293.1                    | 2414.5                        | 121.39                         | 2432.3                    | 2553.7                        | 0.4224                         | 8.0510                    | 8.4734                        |
| 5.0              | 32.87                              | 0.001005                               | 28.185                        |                               | 137.75                         | 2282.1                    | 2419.8                        | 137.75                         | 2423.0                    | 2560.7                        | 0.4762                         | 7.9176                    | 8.3938                        |
| 7.5              | 40.29                              | 0.001008                               | 19.233                        |                               | 168.74                         | 2261.1                    | 2429.8                        | 168.75                         | 2405.3                    | 2574.0                        | 0.5763                         | 7.6738                    | 8.2501                        |
| 10               | 45.81                              | 0.001010                               | 14.670                        |                               | 191.79                         | 2245.4                    | 2437.2                        | 191.81                         | 2392.1                    | 2583.9                        | 0.6492                         | 7.4996                    | 8.1488                        |
| 15               | 53.97                              | 0.001014                               | 10.020                        |                               | 225.93                         | 2222.1                    | 2448.0                        | 225.94                         | 2372.3                    | 2598.3                        | 0.7549                         | 7.2522                    | 8.0071                        |
| 20               | 60.06                              | 0.001017                               | 7.6481                        |                               | 251.40                         | 2204.6                    | 2456.0                        | 251.42                         | 2357.5                    | 2608.9                        | 0.8320                         | 7.0752                    | 7.9073                        |
| 25               | 64.96                              | 0.001020                               | 6.2034                        |                               | 271.93                         | 2190.4                    | 2462.4                        | 271.96                         | 2345.5                    | 2617.5                        | 0.8932                         | 6.9370                    | 7.8302                        |
| 30               | 69.09                              | 0.001022                               | 5.2287                        |                               | 289.24                         | 2178.5                    | 2467.7                        | 289.27                         | 2335.3                    | 2624.6                        | 0.9441                         | 6.8234                    | 7.7675                        |
| 40               | 75.86                              | 0.001026                               | 3.9933                        |                               | 317.58                         | 2158.8                    | 2476.3                        | 317.62                         | 2318.4                    | 2636.1                        | 1.0261                         | 6.6430                    | 7.6691                        |
| 50               | 81.32                              | 0.001030                               | 3.2403                        |                               | 340.49                         | 2142.7                    | 2483.2                        | 340.54                         | 2304.7                    | 2645.2                        | 1.0912                         | 6.5019                    | 7.5931                        |
| 75               | 91.76                              | 0.001037                               | 2.2172                        |                               | 384.36                         | 2111.8                    | 2496.1                        | 384.44                         | 2278.0                    | 2662.4                        | 1.2132                         | 6.2426                    | 7.4558                        |
| 100              | 99.61                              | 0.001043                               | 1.6941                        |                               | 417.40                         | 2088.2                    | 2505.6                        | 417.51                         | 2257.5                    | 2675.0                        | 1.3028                         | 6.0562                    | 7.3589                        |
| 101.325          | 99.97                              | 0.001043                               | 1.6734                        |                               | 418.95                         | 2087.0                    | 2506.0                        | 419.06                         | 2256.5                    | 2675.6                        | 1.3069                         | 6.0476                    | 7.3545                        |
| 125              | 105.97                             | 0.001048                               | 1.3750                        |                               | 444.23                         | 2068.8                    | 2513.0                        | 444.36                         | 2240.6                    | 2684.9                        | 1.3741                         | 5.9100                    | 7.2841                        |
| 150              | 111.35                             | 0.001053                               | 1.1594                        |                               | 466.97                         | 2052.3                    | 2519.2                        | 467.13                         | 2226.0                    | 2693.1                        | 1.4337                         | 5.7894                    | 7.2231                        |
| 175              | 116.04                             | 0.001057                               | 1.0037                        |                               | 486.82                         | 2037.7                    | 2524.5                        | 487.01                         | 2213.1                    | 2700.2                        | 1.4850                         | 5.6865                    | 7.1716                        |
| 200              | 120.21                             | 0.001061                               | 0.88578                       |                               | 504.50                         | 2024.6                    | 2529.1                        | 504.71                         | 2201.6                    | 2706.3                        | 1.5302                         | 5.5968                    | 7.1270                        |
| 225              | 123.97                             | 0.001064                               | 0.79329                       |                               | 520.47                         | 2012.7                    | 2533.2                        | 520.71                         | 2191.0                    | 2711.7                        | 1.5706                         | 5.5171                    | 7.0877                        |
| 250              | 127.41                             | 0.001067                               | 0.71873                       |                               | 535.08                         | 2001.8                    | 2536.8                        | 535.35                         | 2181.2                    | 2716.5                        | 1.6072                         | 5.4453                    | 7.0525                        |
| 275              | 130.58                             | 0.001070                               | 0.65932                       |                               | 548.57                         | 1991.6                    | 2540.1                        | 548.86                         | 2172.0                    | 2720.9                        | 1.6408                         | 5.3800                    | 7.0207                        |
| 300              | 133.52                             | 0.001073                               | 0.60582                       |                               | 561.11                         | 1982.1                    | 2543.2                        | 561.43                         | 2163.5                    | 2724.9                        | 1.6717                         | 5.3200                    | 6.9917                        |
| 325              | 136.27                             | 0.001076                               | 0.56199                       |                               | 572.84                         | 1973.1                    | 2545.9                        | 573.19                         | 2155.4                    | 2728.6                        | 1.7005                         | 5.2645                    | 6.9650                        |
| 350              | 138.86                             | 0.001079                               | 0.52422                       |                               | 583.89                         | 1964.6                    | 2548.5                        | 584.26                         | 2147.7                    | 2732.0                        | 1.7274                         | 5.2128                    | 6.9402                        |
| 375              | 141.30                             | 0.001081                               | 0.49133                       |                               | 594.32                         | 1956.6                    | 2550.9                        | 594.73                         | 2140.4                    | 2735.1                        | 1.7526                         | 5.1645                    | 6.9171                        |
| 400              | 143.61                             | 0.001084                               | 0.46242                       |                               | 604.22                         | 1948.9                    | 2553.1                        | 604.66                         | 2133.4                    | 2738.1                        | 1.7765                         | 5.1191                    | 6.8955                        |
| 450              | 147.90                             | 0.001088                               | 0.41392                       |                               | 622.65                         | 1934.5                    | 2557.1                        | 623.14                         | 2120.3                    | 2743.4                        | 1.8205                         | 5.0356                    | 6.8561                        |
| 500              | 151.83                             | 0.001093                               | 0.37483                       |                               | 639.54                         | 1921.2                    | 2560.7                        | 640.09                         | 2108.0                    | 2748.1                        | 1.8604                         | 4.9603                    | 6.8207                        |
| 550              | 155.46                             | 0.001097                               | 0.34261                       |                               | 655.16                         | 1908.8                    | 2563.9                        | 655.77                         | 2096.6                    | 2752.4                        | 1.8970                         | 4.8916                    | 6.7886                        |
| 600              | 158.83                             | 0.001101                               | 0.31560                       |                               | 669.72                         | 1897.1                    | 2566.8                        | 670.38                         | 2085.8                    | 2756.2                        | 1.9308                         | 4.8285                    | 6.7593                        |
| 650              | 161.98                             | 0.001104                               | 0.29260                       |                               | 683.37                         | 1886.1                    | 2569.4                        | 684.08                         | 2075.5                    | 2759.6                        | 1.9623                         | 4.7699                    | 6.7322                        |
| 700              | 164.95                             | 0.001108                               | 0.27278                       |                               | 696.23                         | 1875.6                    | 2571.8                        | 697.00                         | 2065.8                    | 2762.8                        | 1.9918                         | 4.7153                    | 6.7071                        |
| 750              | 167.75                             | 0.001111                               | 0.25552                       |                               | 708.40                         | 1865.6                    | 2574.0                        | 709.24                         | 2056.4                    | 2765.7                        | 2.0195                         | 4.6642                    | 6.6837                        |

Interpolation: @ P = 230 kPa.

$$v_g = \frac{(230-225)(0.71873-0.79329)}{1250-225} + 0.79329 = 0.778378 \text{ m}^3/\text{kg.}$$

> v<sub>1</sub>.

⇒ mixture.

$$u_f = \frac{(230-225)(0.001067-0.001064)}{25} + 0.001064 = 0.0010646 \text{ kJ/kg.}$$

Find x:  $x = \frac{v - v_f}{v_{fg}} = \frac{v - v_f}{v_g - v_f}$

$$= \frac{0.4140 - 0.0010646}{0.778378 - 0.0010646} = 0.5349.$$

Keep using table A-5 to find u.

$$\left. \begin{array}{l} u_g = \frac{(230-225)(2536.8-2533.2)}{25} + 2533.2 = 2532.48 \text{ kJ/kg.} \\ u_f = \frac{(230-225)(535.08-520.47)}{25} + 520.47 = 523.392 \text{ kJ/kg.} \end{array} \right\} \Rightarrow u(230 \text{ kPa}) = x \cdot u_g + (1-x) u_f$$

$$\Rightarrow u(230 \text{ kPa}) = 0.5349 \times 2532.48 + (1-0.5349) \times 523.392 = 1598.053171 \text{ kJ/kg} \approx 1598.05 \text{ kJ/kg}$$

⇒ The total heat transferred:

$$Q - W = \Delta U = \Delta u \cdot m$$

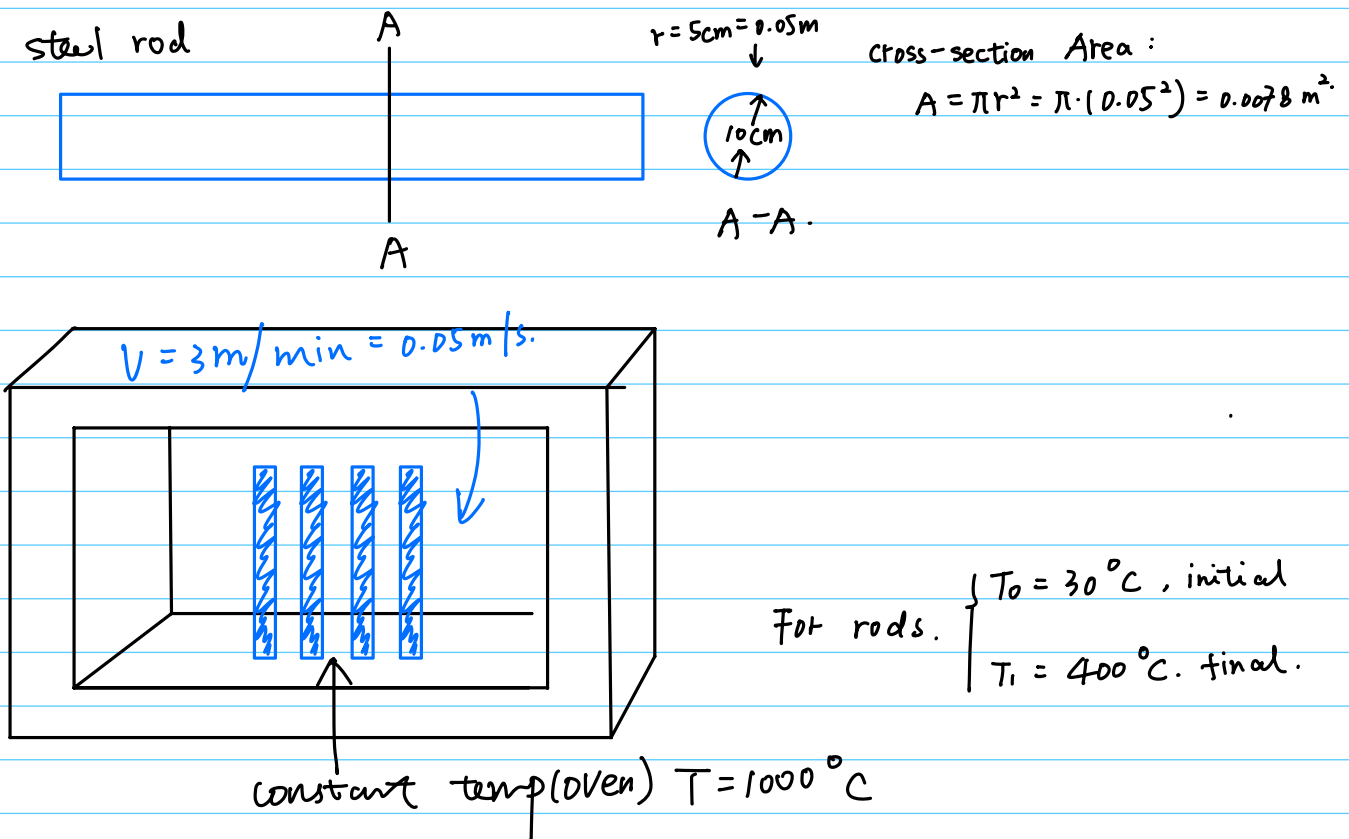
Rearrang :  $Q = \Delta u \cdot m + W$  698.75 kJ.

↑  
 $(1598.05 - 513.44) \text{ kJ/kg} \times 13.28 \text{ kg}$

⇒  $Q = (1598.05 - 513.44) \times 13.28 + 698.75 \approx \underline{15102 \text{ kJ.}}$

Q4.

Long cylindrical steel rods ( $\rho = 7833 \text{ kg/m}^3$  and  $c_p = 0.465 \text{ kJ/kg}\cdot^\circ\text{C}$ ) of 10 cm diameter are heat-treated by drawing them at a velocity of 3 m/min through an oven maintained at  $1000^\circ\text{C}$ . If the rods enter the oven at  $30^\circ\text{C}$  and leave at a mean temperature of  $400^\circ\text{C}$ , determine the rate of heat transfer to the rods in the oven.



The rate of heat transferred to the rods:

specific heat capacity =  $0.465 \text{ kJ/kg}\cdot^\circ\text{C}$

$$\dot{Q} = \dot{m} c_p (T_1 - T_0)$$

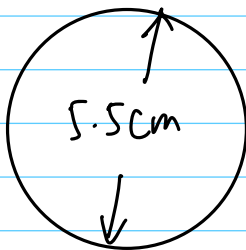
Annotations:   
 $\dot{Q}$  (green arrow) is heat transfer rate.   
 $\dot{m}$  (orange arrow) is mass flow rate.   
 $c_p$  (red arrow) is specific heat capacity.   
 $(T_1 - T_0)$  (blue arrow) is Temperature difference.   
 $T_1 - T_0 = 400 - 30 = 370^\circ\text{C}$

mass flow rate:  $\dot{m} = \rho \cdot A \cdot V = 7833 \times 0.0078 \times 0.05 = 3.05487 \text{ kg/s} \approx 3.05 \text{ kg/s}$

$$\Rightarrow \dot{Q} = 3.05 \times 0.465 \times 370 = 524.7525 \text{ W} \approx 525 \text{ W}$$

Q5.

An ordinary egg can be approximated as a 5.5-cm-diameter sphere. The egg is initially at a uniform temperature of 4°C and is dropped into boiling water at 97°C. Taking the properties of the egg to be  $\rho = 1020 \text{ kg/m}^3$  and  $c_p = 3.32 \text{ kJ/kg} \cdot ^\circ\text{C}$ , determine how much heat is transferred to the egg by the time the average temperature of the egg rises to 90°C.



Heat transferred:

$$Q = m \cdot c_p \cdot \Delta T.$$

$$m = \rho V \quad \& \quad V = \frac{4}{3} \pi r^3.$$

$$r = 2.75 \text{ cm} = 0.0275 \text{ m} \quad \Rightarrow \quad V = \frac{4}{3} \pi (0.0275)^3 = 8.71 \times 10^{-5} \text{ m}^3.$$

$$\text{Given } \rho = 1020 \text{ kg/m}^3, \quad m = \rho V = 8.71 \times 10^{-5} \times 1020 = 0.0888 \text{ kg}.$$

$$c_p = 3.32 \text{ kJ/kg} \cdot ^\circ\text{C}.$$

$$\Delta T = 90^\circ\text{C} - 4^\circ\text{C} = 86^\circ\text{C}$$

$$\Rightarrow Q = 0.0888 \times 3.32 \times 86 = 25.35 \text{ kJ} \approx 25 \text{ kJ}$$